

Electronic Supplementary Information for:

Gold nanoparticles supported into zirconia-ceria mesoporous thin films: a highly active reusable heterogeneous nanocatalyst

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GIXRD characterization

Figure S1 shows the GIXRD and associated Rietveld refinements of a selection of film compositions. The patterns indicate that the walls of the mesopores are crystalline.

Refinements were carried out in terms of a tetragonal cell ($P42/nmc$) although refinement on a cubic cell was also possible. The cell volumes show a linear increase from 68.557 to 73.885 $\text{\AA}^3$ as the Ce content increases confirming that a solid-solution is indeed obtained (Figure S2). This linear increase in cell volume matches very well that reported by Yashima¹ for a series of bulk Ce-substituted tetragonal zirconia powders. Both the change in lattice parameters and the broadening provide strong evidence that Ce(III/IV) has substituted Zr(IV) in the tetragonal nanocrystallite structure. The peak broadening of the principal reflection indicates that crystallite size decreases as Ce increasingly substitutes for Zr.

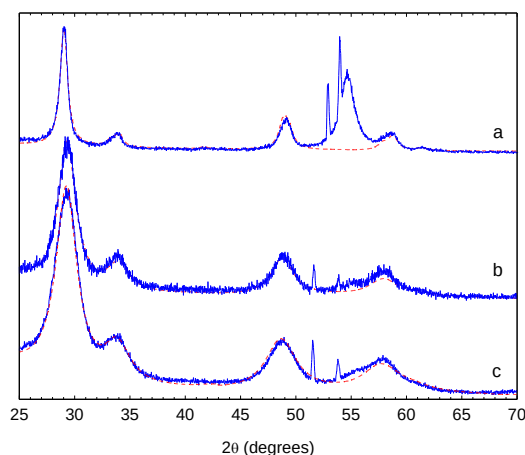


Fig. S1 GIXRD and associated Rietveld refinements of (a) $\text{Zr}_{0.9}\text{Ce}_{0.1}\text{O}_2$, (b) $\text{Zr}_{0.6}\text{Ce}_{0.4}\text{O}_2$ and (c) $\text{Zr}_{0.5}\text{Ce}_{0.5}\text{O}_2$. The two sharp and the broad reflections between 51° and 57° 2θ are due to the silicon substrate.

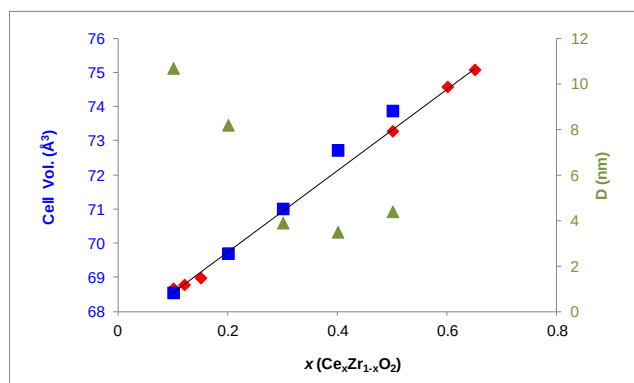


Fig. S2 Values of the cell volume as a function of composition for the thin films (blue squares). Also plotted are the data for bulk materials from the work of Yashima¹ (red diamonds). The green triangles represent the crystallite sizes.

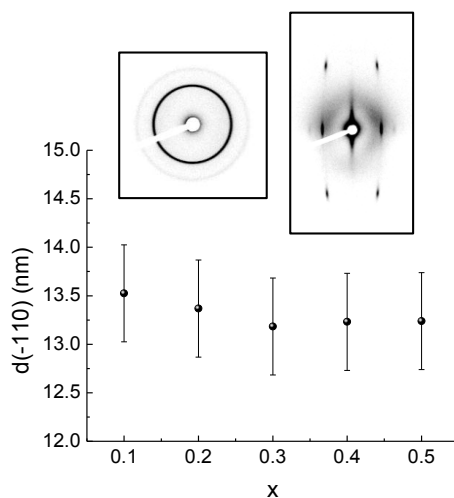


Fig. S3 SAXS data of $\text{Zr}_x\text{Ce}_x\text{O}_2$ mesoporous thin films: interplanar distance $d(-1\ 1\ 0)$ as a function of composition (x). Insets: 2D-SAXS of $\text{Zr}_{0.7}\text{Ce}_{0.3}\text{O}_2$ obtained at 90° (circle) and at 3° showing the typical dots pattern for $Im\bar{3}m$ structure.

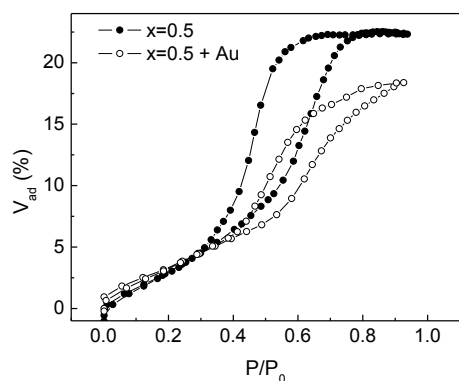


Fig. S4 Accessible volume as a function of the relative water pressure of $\text{Zr}_{0.5}\text{Ce}_{0.5}\text{O}_2$ (full symbols) and $\text{Au}@ \text{Zr}_{0.5}\text{Ce}_{0.5}\text{O}_2$ incorporated after 4 cycles of adsorption-reduction (empty symbols), determined by Environmental ellipsoporosimetry.

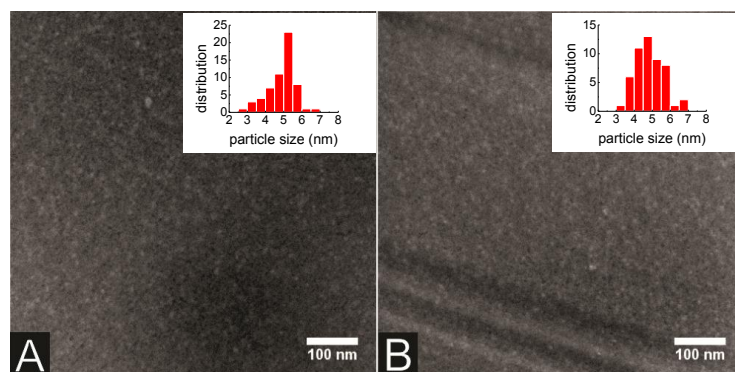


Fig. S5: FESEM of a $\text{Zr}_{0.5}\text{Ce}_{0.5}\text{O}_2$ film before and after use in 4NIP reduction. The measured size of the AuNPs is $4.8 \pm 0.8 \text{ nm}$ in both cases. The scratch marks in figure B are due to TEM sample preparation.

Electrostatically adsorbed AuNPs

In order to establish the durability and reusability of gold nanoparticles supported on a bare substrate, electrostatically adsorbed AuNPs onto soda-lime glass were prepared. Turkevich method² was used to produce monodispersed 15nm-sized negatively charged AuNPs. They

were attached to a substrate functionalized with positively charged polymer PDDA, as previously reported by Angelomé et al.³ This attached AuNPs were tested in the reduction of 4NIP in the presence of BH_4^- . In this case, the reaction proceeds even after the glass support is removed, due to AuNPs detachment.

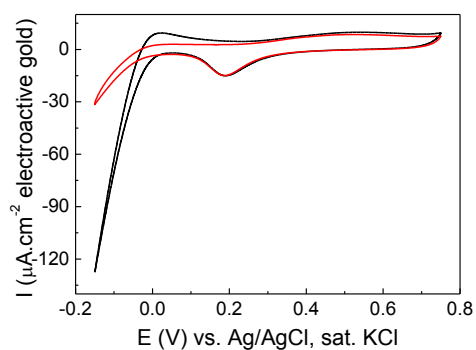


Fig. S6 Cyclic voltammograms of AuNps@ $\text{Zr}_{0.5}\text{Ce}_{0.5}\text{O}_2$ /ITO electrodes in 0.01 M KOH (red line) and the reduction of oxygen in 0.01M KOH saturated in O_2 (black line).

1. Yashima, M.; Hirose, T.; Katano, S.; Suzuki, Y.; Kakihana, M.; Yoshimura, M., Structural Changes of ZrO_2 - CeO_2 Solid Solutions Around the Monoclinic-Tetragonal Phase Boundary. *Phys. Rev. B* **1995**, *51*, 8018-8025.
2. Enustun, B. V.; Turkevich, J., Coagulation of Colloidal Gold. *J. Am. Chem. Soc.* **1963**, *85*, 3317-3328.
3. Angelomé, P. C.; Liz-Marzán, L. M., Monitoring Solvent Evaporation from Thin Films by Localized Surface Plasmon Resonance Shifts. *J. Phys. Chem. C* **2010**, *114*, 18379-18383.